

# Duration of antibody protection project

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## Abstract

How much longitudinal antibody data must a vaccine trial accumulate before it can make a credible statement about the duration of protection? At a planned review, this becomes a practical design question: is the evidence sufficient for durability follow-up to conclude, or is further observation still likely to change the answer? This workshop project will approach that question through simulation and model-based extrapolation. The working application is COVID-19 vaccination, for which both antibody-kinetics models and quantitative links between neutralisation and protection are available. The choices below are starting points for discussion, not a fixed analysis protocol.

## 1 Project question and aims

Consider a vaccine trial, or a long-term extension, that measures an antibody marker repeatedly after vaccination. Antibody levels often fall quickly at first and more slowly later. We would like to estimate when the marker will fall below a protection-relevant level, even when that crossing occurs beyond the observed follow-up. Immune correlates require careful terminology—a correlate can predict protection without being its sole cause—but they can still support useful quantitative predictions (Plotkin and Gilbert 2012; Earle et al. 2021).

The central question is therefore:

*At a planned trial review, how long must participants have been followed, and how informative must their longitudinal measurements be, before the duration of antibody-associated protection can be estimated credibly enough to decide whether further durability follow-up is needed?*

The project has four linked aims:

1. describe plausible short- and long-term antibody trajectories;
2. quantify how much extrapolation is being asked of a fitted model;
3. understand when apparently similar fits imply very different durations of protection;
4. translate those findings into practical guidance on the duration and timing of trial follow-up.

The distinction between concluding durability follow-up and stopping all trial activity is important: safety surveillance, clinical endpoints and regulatory commitments may independently require continued follow-up. The present project concerns the information available for the durability question.

This decision problem is clearest when protection is expected to last for years. The pivotal Shingrix trials followed participants for roughly three to four years, while their long-term extension ultimately assessed efficacy and immunogenicity through 11 years after vaccination (Strezova et al. 2025). Such studies show the value of long follow-up, but also motivate asking at which earlier review points the accumulated marker data can already support a reliable conclusion, and when additional observations are still needed to resolve the tail.

COVID-19 provides a well-developed starting point: neutralisation is predictive of protection, and published work connects titres, variants, boosting and vaccine effectiveness (Khoury et al. 2021; Cromer et al. 2022; Gilbert et al. 2022; Feng et al. 2021). Whether the same framework transfers to other pathogens, assays and types of immune marker is deliberately left open for the workshop.

## 2 What do we mean by duration of protection?

Let  $c_i(t)$  denote the latent antibody concentration for participant  $i$  at time  $t$ , and let  $c_{\text{thr}}$  be a protection-relevant threshold on the same assay scale. A natural individual-level target is the first threshold-crossing time

$$T_i^* = \inf\{t \geq 0 : c_i(t) \leq c_{\text{thr}}\}. \quad (1)$$

This quantity is a model-based estimand, not a directly observed biological expiry date. It can be undefined if the trajectory approaches a plateau above the threshold, and it depends on the endpoint and threshold definition.

For the initial COVID-19 application, one option is to anchor  $c_{\text{thr}}$  to the neutralisation level associated with 50% protection in the logistic relationship estimated by Khoury et al. (2021). The same fitted antibody curve could instead be passed through the full titre-to-protection relationship to obtain a protection curve  $E_i(t)$ . This makes explicit that “antibody duration” and “protection duration” are related but not interchangeable.

Heterogeneity creates a second distinction. The distribution of individual  $T_i^*$  values is not equivalent to the threshold-crossing time of a population-average antibody curve. A population target might instead be the time at which the fraction of people still above the threshold falls below a chosen level. The workshop should keep individual and population questions separate.

## 3 Candidate trajectory models

The candidate set is intended to expose different assumptions about the unobserved tail. We suggest the biphasic exponential as a common reference model for the first analyses, while retaining the other families as substantive alternatives.

### Single exponential

$$c(t) = c_0 e^{-kt}. \quad (2)$$

This is the simplest reference model. The log concentration has a constant slope and  $T^* = k^{-1} \log(c_0/c_{\text{thr}})$  when  $c_0 > c_{\text{thr}}$ . It can be identified from relatively short follow-up if the trajectory is truly exponential, but it extrapolates the early slope indefinitely when that assumption is wrong. Single-half-life summaries are common in analyses of COVID-19 neutralisation waning (Khoury et al. 2021; Cromer et al. 2022).

## Biphasic exponential (suggested default)

Biphasic structures have been used successfully to describe vaccine-antibody kinetics in established analyses, representing rapid initial contraction followed by a slower persistent component (M. T. White et al. 2015; Hogan et al. 2023). We therefore suggest this family as the initial reference model: its parameters have a useful biological interpretation, its long-term driver is explicit, and a published COVID-19 calibration is available. This is a practical starting point for comparison, rather than a claim that biphasic decay will be preferred for every dataset. We use the smooth parameterisation of Hogan et al. (2023) from the outset:

$$c(t) = c_0\{w_1e^{-k_1t} + w_2e^{-k_2t}\}, \quad k_r = \frac{\log 2}{h_r}, \quad r \in \{1, 2\}, \quad h_1 < h_2, \quad (3)$$

$$D = e^{-k_1t_s} + e^{-k_2t_s}, \quad w_1 = \frac{e^{-k_2t_s}}{D}, \quad w_2 = \frac{e^{-k_1t_s}}{D}. \quad (4)$$

The fast component can represent contraction of short-lived antibody-secreting cells and the slow component a longer-lived population. The weights ensure  $c(0) = c_0$  and make the two weighted components equal at switching time  $t_s$ . The slow half-life  $h_2$  governs long-term duration but may be weakly identified while the fast component dominates.

## Power-law decay

$$c(t) = c_0(1 + \alpha t)^{-\beta}. \quad (5)$$

This curve decelerates continuously. It can resemble an exponential over a short interval while implying a much longer tail. Long-term antibody data provide biological motivation for considering decelerating decay rather than a constant half-life (Amanna et al. 2007).

## Plateau or set-point

$$c(t) = A_\infty + (c_0 - A_\infty)e^{-kt}. \quad (6)$$

Here the long-term level is non-zero. A crossing exists only when  $A_\infty < c_{\text{thr}}$ . Short data may not distinguish a genuine plateau from a very slow decline, making “never crosses” an informative possibility rather than a numerical failure. Long-term measurements showing durable antibody levels motivate this alternative (Amanna et al. 2007).

## Mechanistic antibody-production model

A richer generating model can represent short- and long-lived antibody-secreting cells and antibody clearance separately:

$$\frac{dP_s}{dt} = -\delta_s P_s, \quad \frac{dP_l}{dt} = -\delta_l P_l, \quad \frac{dc}{dt} = \rho_s P_s + \rho_l P_l - \delta_c c. \quad (7)$$

When antibody clears quickly relative to the cells, this approaches biphasic decay; added memory or replenishment can generate an even slower tail. Related models have been developed for plasma cell dynamics and vaccine responses (Andraud et al. 2012; Pasin et al. 2019). Such a model may be most useful as a deliberately richer data-generating mechanism rather than as the default fitted model.

A complementary later-stage generator could use the semi-mechanistic model developed across a series of COVID-19 studies (Nishiyama et al. 2023; Nakamura et al. 2024; Park et al. 2025; Hart et al. 2025). It links a decaying mRNA stimulus to delayed, nonlinear antibody production and

antibody loss, and can represent boosting, infection history and between-participant variation. Public code and derived parameter estimates make it a practical calibration resource. Once vaccine stimulation has vanished, however, its long-term tail tends towards a single exponential, so it is better suited to testing dose-response dynamics and heterogeneous trajectories than to replacing the richer slow-tail generator above.

## 4 Observation model and biphasic population parameterisation

Observed assays are naturally treated as noisy measurements of a latent positive curve. A simple starting point is multiplicative log-normal error,

$$\log y_{ij} = \log c_i(t_{ij}) + \epsilon_{ij}, \quad \epsilon_{ij} \sim N(0, \sigma_{\log}^2). \quad (8)$$

Lower and upper quantification limits can be represented as censoring rather than replacing censored observations by exact boundary values. This observation model can be used with any of the candidate trajectories in Section 3.

For the suggested biphasic reference model, between-participant variation can be introduced through correlated log-normal random effects on peak level and the slow decay rate, for example

$$\begin{aligned} c_{0i} &= c_0 e^{\eta_{0i}}, & k_{2i} &= k_2 e^{\eta_{ki}}, & (9) \\ \begin{pmatrix} \eta_{0i} \\ \eta_{ki} \end{pmatrix} &\sim N_2 \left[ \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \omega_0^2 & \rho_{0k} \omega_0 \omega_k \\ \rho_{0k} \omega_0 \omega_k & \omega_k^2 \end{pmatrix} \right]. & (10) \end{aligned}$$

This permits both individual and population targets and asks whether additional participants compensate for sparse information within each trajectory. Population mechanistic models have used this type of structure in vaccine applications (Pasin et al. 2019).

Table 1 gives tentative values for this biphasic baseline around which to begin discussion. They are calibration aids, not universal biological constants. In particular, assay units and protection thresholds are not automatically portable between studies. The source/basis column distinguishes published estimates from workshop modelling choices.

Table 1: Tentative starting values for the suggested biphasic COVID-19 baseline.

| Quantity                               | Starting value | Source / basis                   | Interpretation / possible variation                                 |
|----------------------------------------|----------------|----------------------------------|---------------------------------------------------------------------|
| Early level $c_0$                      | 5000 BAU/mL    | Workshop scale choice            | Illustrative assay scale; consider 3000–8000                        |
| Log-error SD $\sigma_{\log}$           | 0.20           | Workshop noise choice            | Explore lower and higher assay variability                          |
| Fast half-life $h_1$                   | 35 days        | Hogan et al. (2023)              | Early contraction; tentative range 25–60 days                       |
| Slow half-life $h_2$                   | 581 days       | Hogan et al. (2023)              | Highest-priority tail sensitivity; consider 250–900 days            |
| Switching time $t_s$                   | 75 days        | Hogan et al. (2023)              | Equal weighted components at this time                              |
| Weights $w_1 : w_2$                    | about 80:20    | Derived from Hogan et al. (2023) | From $h_1$ , $h_2$ and $t_s$ in Equation 4                          |
| Protection anchor $c_{\text{thr}}$     | 20.2% / 3.0%   | Khoury et al. (2021)             | Convalescent neutralisation for 50% symptomatic / severe protection |
| Peak / rate log-SD $\omega_0/\omega_k$ | 0.30 / 0.25    | Workshop heterogeneity choice    | Moderate between-participant variation                              |
| Peak-rate correlation $\rho_{0k}$      | −0.30          | Workshop correlation choice      | Include zero as a comparison                                        |

The workshop choices in the table are deliberately round simulation defaults, not estimates attributed to the cited studies. The workshop may choose a different assay normalisation or tail calibration before any headline analysis.

## 5 A possible analysis programme

A simulation-first programme would create a full longitudinal dataset from a known trajectory, then reveal the portions available at successive planned trial reviews. Each candidate model could be fitted to the data available at each review and extrapolated to  $T^*$ . Because the generating truth is known, the exercise can separate uncertainty from confident misspecification and ask when a decision to conclude durability follow-up would have been well supported.

A natural progression is:

1. establish a transparent homogeneous baseline in which a biphasic model is fitted to biphasic data;
2. reveal successively longer follow-up windows and examine when conclusions about duration become stable, or only appear stable;
3. fit all candidate families to the same truncated data to expose disagreement in the tail;
4. generate data from richer mechanistic models and ask whether simple fitted models remain useful;
5. introduce participant heterogeneity and compare individual with population duration;
6. if useful, map antibody trajectories to symptomatic and severe-disease protection or compare the framework with an open real dataset.

Points for discussion include which duration is clinically meaningful; how strongly to privilege COVID-19; which model families and parameterisations are worth pursuing; how inference and study design should be handled; what evidence should be sufficient to conclude durability follow-up; which comparisons would be most informative; and what degree of extrapolation would be acceptable to a trial designer.

## 6 How the literature informs the project

**Plotkin and Gilbert (2012).** This short framework gives us the vocabulary needed to distinguish a statistical correlate, a mechanistic correlate and a surrogate endpoint, helping us avoid interpreting a protection-relevant antibody marker as the whole immune mechanism (Plotkin and Gilbert 2012).

**Khoury et al. (2021).** This is the main quantitative bridge from neutralisation titre to protection: it supplies the logistic relationship and relative 50%-protection anchors that can be used to turn a fitted antibody curve into an endpoint-specific protection curve (Khoury et al. 2021).

**Cromer et al. (2022).** The meta-analysis extends the titre-to-protection framework across variants and boosting, reminding us that starting titre and antigenic change can alter protection duration even when the underlying decay process is unchanged (Cromer et al. 2022).

**Gilbert et al. (2022).** The mRNA-1273 efficacy-trial analysis provides trial-level evidence that neutralisation is a graded correlate of risk, supporting continuous marker models rather than only a binary laboratory cut-off (Gilbert et al. 2022).

**Feng et al. (2021).** This analysis of ChAdOx1 nCoV-19 similarly supports a continuous relationship between immune markers and symptomatic or asymptomatic infection, and cautions against treating any single threshold as universal (Feng et al. 2021).

**Earle et al. (2021).** The cross-vaccine synthesis supplies broader evidence that antibody levels can function as a protective correlate for COVID-19 vaccines, strengthening the case for the marker-based starting point while not establishing causality by itself (Earle et al. 2021).

**Hogan et al. (2023).** This model-based study provides the tentative COVID-19 biphasic calibration—including fast and slow half-lives and a switching time—and illustrates how immune kinetics can be connected to longer-term effectiveness against variants (Hogan et al. 2023).

**Nishiyama et al. (2023).** This study extends an mRNA-stimulus antibody model to a booster dose and infection history, showing how repeated immune exposures can alter response duration and providing a compact semi-mechanistic alternative for later simulations (Nishiyama et al. 2023).

**Nakamura et al. (2024).** Dense measurements from a small calibration cohort and sparse measurements from more than 2,400 participants are combined to reconstruct heterogeneous individual trajectories after primary vaccination. The fitted parameter distributions are a useful COVID-19 calibration source, although the underlying clinical dataset is access-restricted (Nakamura et al. 2024).

**Park et al. (2025).** Longitudinal trajectories after primary and booster vaccination are stratified into durable, vulnerable and rapid-declining response patterns, with rapid decline linked to earlier subsequent infection. This makes heterogeneity clinically relevant rather than merely a statistical complication (Park et al. 2025).

**Hart et al. (2025).** The same individual-level antibody model is linked to susceptibility and population outbreak risk. Its open code and derived parameter estimates provide an immediately usable resource for simulation and calibration (Hart et al. 2025).

**White et al. (2015).** The RTS,S malaria analysis is the closest published analogue to the project’s intended chain: it uses biphasic antibody kinetics and links the resulting waning marker to duration of vaccine efficacy (M. T. White et al. 2015).

**Strezova et al. (2025).** The final Shingrix long-term follow-up analysis demonstrates the practical scale of the design problem: an extension of pivotal phase 3 trials followed more than 7,000 participants and assessed protection through 11 years. It provides a concrete setting in which the value of additional durability data must be weighed against prolonged follow-up (Strezova et al. 2025).

**Andraud et al. (2012).** The nested plasma-cell and antibody models provide a clear mechanistic template for generating trajectories on several biological time scales and for asking when a simpler decay curve is an adequate approximation (Andraud et al. 2012).

**Pasin et al. (2019).** The Ebola prime-boost study demonstrates how mechanistic antibody dynamics and between-participant variation can be estimated together, making it a useful model for the hierarchical phase of this project (Pasin et al. 2019).

**Amanna et al. (2007).** Long-term measurements across antigens show that antibody durability varies greatly and need not follow a single constant half-life, motivating the power-law and plateau alternatives and explicit attention to the tail (Amanna et al. 2007).

## Selected literature

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